Boolean algebra, De Morgan’s Law, Logic Gates

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What is Boolean Algebra?

- Boolean algebra is a branch of mathematics that deals with variables that have two possible values: true (1) and false (0)
- It was introduced by George Boole in 1854 in his book "An Investigation of the Laws of Thought."

- Boolean algebra is the foundation of digital logic and computer circuits.

Applications of Boolean Algebra

- Digital logic design
- Computer architecture
- Circuit design
- Programming (conditional statements, loops)
- Search algorithms and database queries

Boolean Variables and Operations

- Variables in Boolean algebra can take only two values: 0 (false) or 1 (true).
- Example: Let A be a Boolean variable. A can be either 0 or 1.

- **AND ($\cdot$)**: The output is true only if all inputs are true.
  - $A \cdot B = 1$ if and only if $A = 1$ and $B = 1$.
- **OR ($+$)**: The output is true if at least one input is true.
  - $A + B = 1$ if $A = 1$ or $B = 1$ or both.
- **NOT ($\overline{A}$)**: The output is the inverse of the input.
  - $\overline{A} = 1$ if $A = 0$ and vice versa.

Truth Tables

- A truth table is a mathematical table used to represent the output of a Boolean function for all possible combinations of inputs.
- Example: Truth table for AND, OR, and NOT operations.

NOT

x	x'
0	1
1	0

AND

x	y	xy
0	0	0
0	1	0
1	0	0
1	1	1

OR

x	y	$x+y$
0	0	0
0	1	1
1	0	1
1	1	1

Basic Laws

- **Identity Laws**:
  - $A + 0 = A$
  - $A \cdot 1 = A$
- **Domination Laws**:
  - $A + 1 = 1$
  - $A \cdot 0 = 0$
- **Complement Laws**:
  - $A + \overline{A} = 1$
  - $A \cdot \overline{A} = 0$

Commutative, Associative, and Distributive Laws

- **Commutative Laws**:
  - $A + B = B + A$
  - $A \cdot B = B \cdot A$
- **Associative Laws**:
  - $A + (B + C) = (A + B) + C$
  - $A \cdot (B \cdot C) = (A \cdot B) \cdot C$
- **Distributive Laws**:
  - $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$
  - $A + (B \cdot C) = (A + B) \cdot (A + C)$

4. Simplification of Boolean Expressions (15 minutes)

4.1 Algebraic Simplification

- Use the laws of Boolean algebra to simplify expressions.
- Example: Simplify $A \cdot (A + B)$
  - $A \cdot (A + B) = A \cdot A + A \cdot B = A + A \cdot B = A \cdot (1 + B) = A \cdot 1 = A$

4.2 Karnaugh Maps (Brief Introduction)

- A graphical method for simplifying Boolean expressions.
- Useful for minimizing the number of logic gates in a circuit.

Table 4-1 Basic rules of Boolean algebra.

1. $A + 0 = A$	**7.** $A \cdot A = A$
2. $A + 1 = 1$	**8.** $A \cdot \overline{A} = 0$
3. $A \cdot 0 = 0$	**9.** $\overline{\overline{A}} = A$
4. $A \cdot 1 = A$	**10.** $A + AB = A$
5. $A + A = A$	**11.** $A + \overline{A}B = A + B$
6. $A + \overline{A} = 1$	**12.** $(A + B)(A + C) = A + BC$

A, B, or *C* can represent a single variable or a combination of variables.

Problem 1: $A + \overline{A}B = A + B$

This rule can be proved as follows:

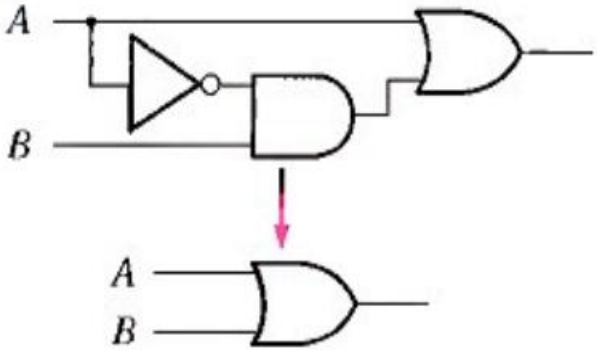
$A + \overline{A}B = (A + AB) + \overline{A}B$	Rule 10: $A = A + AB$
$= (AA + AB) + \overline{A}B$	Rule 7: $A = AA$
$= AA + AB + A\overline{A} + \overline{A}B$	Rule 8: adding $A\overline{A} = 0$
$= (A + \overline{A})(A + B)$	Factoring
$= 1.\ (A + B)$	Rule 6: $A + \overline{A} = 1$
$= A + B$	Rule 4: drop the 1

The proof is shown in Table 4-3, which shows the truth table and the resulting logic circuit simplification.

Table 4-3

A	B	$\overline{A}B$	$A + \overline{A}B$	$A + B$
0	0	0	0	0
0	1	1	1	1
1	0	0	1	1
1	1	0	1	1

equal



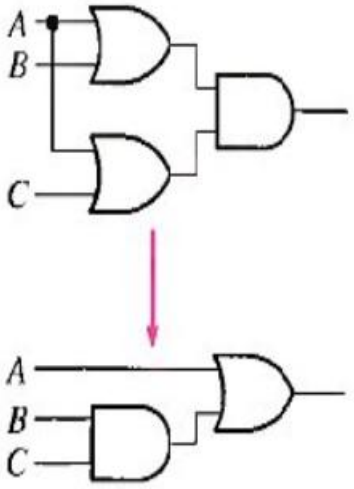
Problem 2: $(A + B)(A + C) = A + BC$

This rule can be proved as follows:

$(A + B)(A + C) = AA + AC + AB + BC$	Distributive law
$= A + AC + AB + BC$	Rule 7: $AA = A$
$= A(1 + C) + AB + BC$	Rule 2: $1 + C = 1$
$= A.\ 1 + AB + BC$	Factoring (distributive law)
$= A(1 + B) + BC$	Rule 2: $1 + B = 1$
$= A.\ 1 + BC$	Rule 4: $A\ .\ 1 = A$
$= A + BC$	

A	B	C	A + B	A + C	(A + B)(A + C)	BC	A + BC
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	1	1	1	1
1	0	0	1	1	1	0	1
1	0	1	1	1	1	0	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

equal



De Morgan's Theorems

- De Morgan, a mathematician who knew Boole, proposed two theorems that are an important part of Boolean algebra.

- In practical terms. De Morgan's theorems provide mathematical verification of the equivalency of the NAND and negative-OR gates and the equivalency of the NOR and negative-AND gates.

De Morgan's Theorems

1. The complement of a product of variables is equal to the sum of the complements of the variables

The formula for expressing this theorem for two variables is

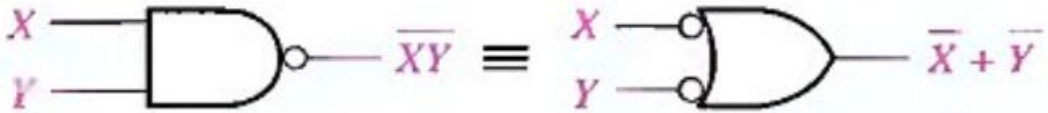
$$\overline{XY} = \overline{X} + \overline{Y}$$

De Morgan's Theorems

2. The complement of a sum of variables is equal to the product of the complements of the variables.

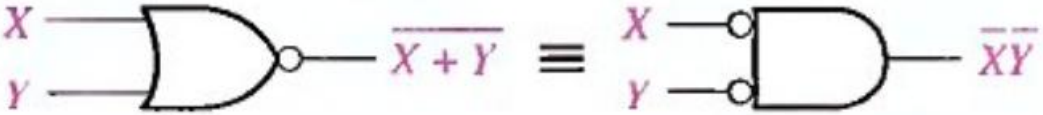
The formula for expressing this theorem for two variables is

$$\overline{X + Y} = \overline{X}\,\overline{Y}$$



NAND Negative-OR

Inputs		Output	
X	Y	$\overline{XY}$	$\overline{X} + \overline{Y}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0



NOR Negative-AND

Inputs		Output	
X	Y	$\overline{X + Y}$	$\overline{X}\,\overline{Y}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

Example

Apply DeMorgan's theorems to the expressions $\overline{XYZ}$ and $\overline{X + Y + z}$.

$$\overline{XYZ} = \overline{X} + \overline{Y} + \overline{Z}$$

$$\overline{X + y + Z} = \overline{X}\,\overline{Y}\,\overline{Z}$$

Example

Apply DeMorgan's theorems to the expressions $\overline{WXYZ}$ and $\overline{W + X + y + z}$.

$$WXYZ = \overline{W} + \overline{X} + \overline{y} + \overline{Z}$$

$$\overline{W + X + y + Z} = \overline{W}\,\overline{X}\,\overline{Y}\,\overline{Z}$$

Applying DeMorgan's Theorems

The following procedure illustrates the application of DeMorgan's theorems and Boolean algebra to the specific expression

$$\overline{\overline{A + B\overline{C}} + D(\overline{E + \overline{F}})}$$

Step 1. Identify the terms to which you can apply DeMorgan's theorems, and think of each term as a single variable. Let $\overline{A + \overline{BC}} = X$ and $D\overline{(E + \overline{F})} = Y$.

Step 2. Since $\overline{X + Y} = \overline{X}\,\overline{Y}$,

$$\overline{\overline{A + B\overline{C}} + D(\overline{E + \overline{F}})} = \overline{\overline{(A + B\overline{C})}}\;\overline{(D\overline{(E + \overline{F})})}$$

Step 3. Use rule 9 ($A = \overline{\overline{A}}$) to cancel the double bars over the left term (this is not part of DeMorgan's theorem).

$$\overline{\overline{(A + B\overline{C})}}\ \overline{(D\overline{\overline{(E + \overline{F})}})} = (A + B\overline{C})\overline{(D\overline{\overline{(E + \overline{F})}})}$$

Step 4. Applying DeMorgan's theorem to the second term,

$$(A + B\overline{C})\overline{(D\overline{\overline{(E + \overline{F})}})} = (A + B\overline{C})(\overline{D} + \overline{\overline{(E + \overline{F})}})$$

<u>Step 5</u>. Use rule 9 ($A = \overline{\overline{A}}$) to cancel the double bars over the $E + \overline{F}$ part of the term.

$$(A + B\overline{C})(\overline{D} + \overline{\overline{E + \overline{F}}}) = (A + B\overline{C})(\overline{D} + E + \overline{F})$$

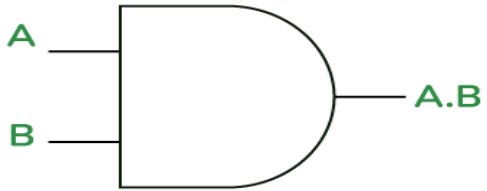
Logic Gates

- A logic gates are an electronic circuit that are designed by using electrical components like diodes, transistors, resistors, and more.
- It is used to perform logical operations based on the inputs provided to it and gives logical output that can be either high(1) or low(0).
- The operation of logic gates is based on the Boolean algebra or mathematics.
- Logic gate founds its uses in our day to day basis such as in the architecture of our telephone, laptops, tablets an memory devices.

AND Gate

An AND gate is used to perform logical Multiplication of binary input. The Output state of the AND gate will be high(1) if both the input are high(1) ,else the output state will be low(0) if any of the input is low(0).

2- Input AND Gate

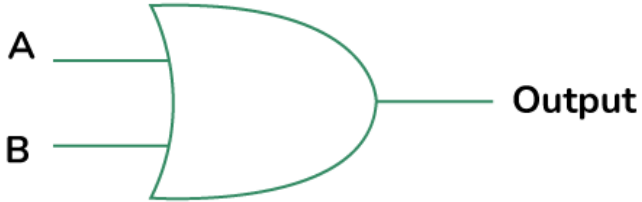


Truth Table

A (Input 1)	B (Input 2)	X = (A.B)
0	0	0
0	1	0
1	0	0
1	1	1

OR GATE is most widely used digital logic circuit. The output state of OR gate will be high i.e.,(1) if any of the input state is high or 1, else output state will be low i.e., 0.

2-Input OR Gate

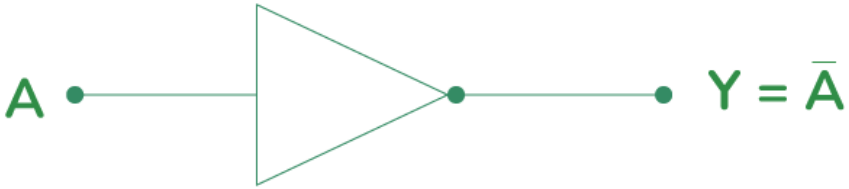


Truth Table

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate

In digital electronics, the NOT gate is one of the basic logic gate having only a single input and a single output. It is also known as inverter or inverting buffer. When the input signal is “low” the output signal is “high” and vice-versa.

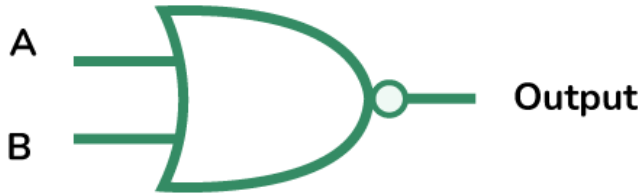


Truth Table

A (Input)	Y = Ā (Output)
0	1
1	0

The NOR gate is the type of universal logic gate. It takes two or more inputs and gives only one output. The output state of the NOR gate will be high(1) when all the inputs are low(0). NOR gate returns the complement result of the OR gate. It is basically a combination of two basic logic gates i.e., OR gate and NOT gate.

2- Input NOR Gate

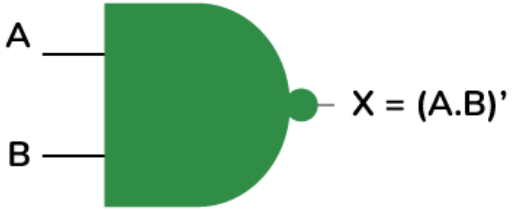


Truth Table

Input A	Input B	0 = (A + B)'
0	0	1
0	1	0
1	0	0
1	1	0

2-Input NAND Gate

The NAND Gate is another type of Universal logic gate. The NAND gate or “Not AND” is the combination of two basic logic gates AND gate and the NOT gate connected in series. It takes two or more inputs and gives only one output. The output of the NAND gate will give result high(1) when either of its input is high(1) or both of its input are low(0). In simple, it performs the inverted operation of AND gate.



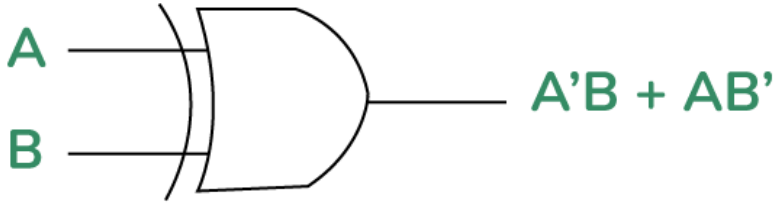
Truth Table

Input A	Input B	X = (A.B)’
0	0	1
0	1	1
1	0	1
1	1	0

In digital electronics, there is a specially designed logic gate named, XOR gate, which is used in digital circuits to perform modulo sum. It is also referred to as Exclusive OR gate or Ex-OR gate. it is used extensively in arithmetic logic circuits., logic comparators and error detection circuits. The XOR gate can take only two inputs at a time and give an output. The output of the XOR gate is high(1) only when its two inputs are dissimilar i.e., if one of them is low(0) then other one will be high(1).

XOR Gate

$$Y = (A \oplus B)$$



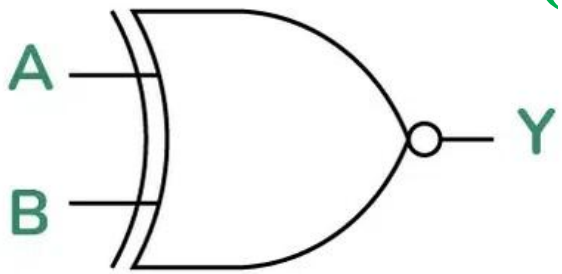
Truth Table

A (Input 1)	B (Input 2)	X = A'B + AB'
0	0	0
0	1	1
1	0	1
1	1	0

The XNOR is the combination of XOR gate and NOT gate. The output of the XNOR gate is high(1) when both the inputs are high(1) or low(0). In other words, the output of the XNOR gate is high(1) when both the inputs are the same. the XNOR gate can be sometimes be called as Equivalence gate. In simple words, The XNOR gate is the complement of the XOR gate.

What is XNOR Gate?

$$Y = (A \oplus B)'$$



$$Y = (\overline{A \oplus B}) = (A.B + \overline{A}.\overline{B})$$

Truth Table

Input A	Input B	Output
0	0	1
0	1	0
1	0	0
1	1	1